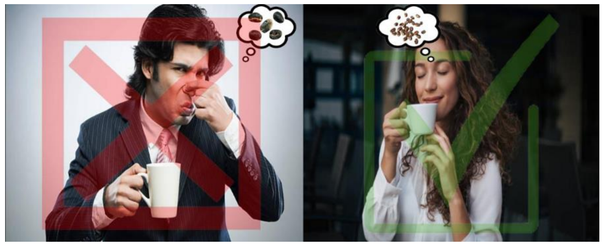
Defect Prediction in Arabica Beans

INTRODUCTION

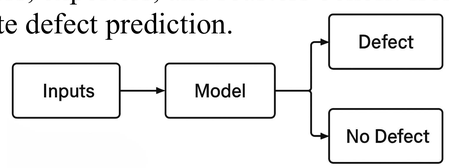
Problem



- Identify factors that drive *defective* Arabica coffee beans.
- Predict defect presence at the batch level.

Motivation

- Quality shifts buyer decisions.
- Producers, exporters, and roasters benefit from fast, accurate defect prediction.



Solution

- Dataset: 8,446 datapoints.
- Manual review: is slow, inconsistent, and error-prone. Thus, we Performed these test computationally
- The model uses the various inputs (combinations of instances of attributes) to learn defect patterns.
- Target performance: performance ≥ 0.8 AUROC & AUPRC

Related Work

Decision-tree analysis of coffee defects: Rizkya, I., Syahputri, K., Sari, R.M., Christin, S. *Analysis of Defective Causes in Coffee Product Using Decision Tree Approach.* [1]

Their problem

- Examining defects and defect causes occurring in coffee bean processing

Their approach

- Using cause and effect diagrams, they used decision tree tools to find each specific cause

Their performance

- They found out they needed more training, observation and controlling at each station

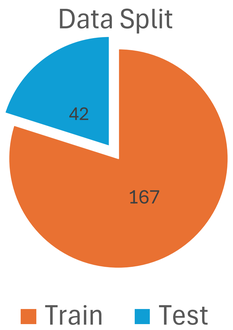
how it compares to us

- Both used decision tree tools to find out what causes defects in processing beans

METHODOLOGY

Data

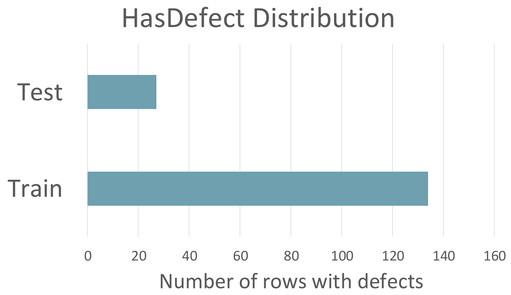
- Source: Coffee Quality Institute (scraped).
- 206 batches , 41 attributes;
- **binary classification** (Defect vs No Defect)
- Train/Test: stratified split 80/20



Attribute	Type
ID	Nominal
CountryOfOrigin	Nominal
LotNumber	Nominal
Company	Nominal
Altitude	Numeric
Region	Nominal
Producer	Nominal
NumberOfBags	Numeric
BagWeight	Numeric
HarvestYear	Nominal
Quakers	Numeric
Color	Nominal
CertificationAddress	Nominal
HasError / HasDefect	Nominal

Pre-processing

- Removed redundant fields.
- Deleted non-ASCII text; cleaned inconsistent strings.
- Merged CategoryOneDefects + CategoryTwoDefects → **HasDefect** (Binary Class Attribute).
- Checked missing/outliers; attributes with 75% + missing or fully uniform were removed



Models

- **KNN**: distance-based neighbor voting.
- **Naïve Bayes**: probabilistic conditional-independence model.
- **SVM**: margin-based linear/nonlinear separator.
- **RandomForest**: tree ensemble with aggregated voting.
- **AdaBoost**: boosting weak learners via reweighting.
- **ZeroR**: majority-class baseline.
- **OneR**: single-attribute rule baseline.

Implementation

- Tools: WEKA (modeling), Python + Excel (cleaning).
- Performed preprocessing, feature ranking, tuning, and evaluation in standard ML workflow.
- Used AI assistance for refinement; final decisions validated manually.

Validation:

- 10-fold cross-validation.

Hyperparameters tuned:

- **KNN**: k = 1–10,
- **Naïve Bayes**: none,
- **SVM:** random seed 1–5
- **AdaBoost:**random seed 1–5
- **RandomForest**: random seed 1–5

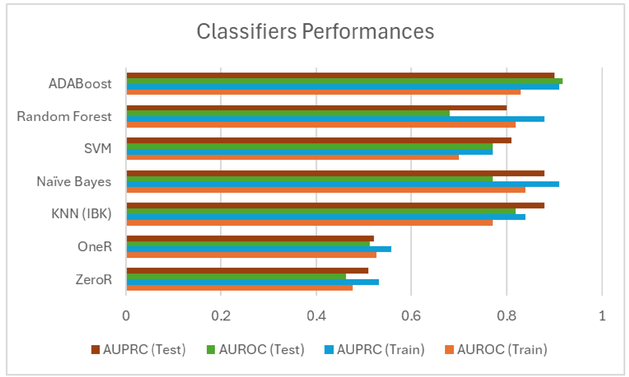
Attribute selection:

- filter-based information-gain ranking

Performance Metrics:

- AUROC: Area under receiver operating characteristic
- AUPRC: Area under precision-recall curve

RESULTS



Adaboost performed best overall

Top attributes used by Adaboost:

- Quakers: almost defective beans
- LotNumber: where its grown
- NumberOfBags: # of bags
- Altitude: altitude of where it grew
- CertificationAddress: unknown

AdaBoost Performance on the test set

Actual \ predicted	TRUE	FALSE
TRUE	26	1
FALSE	9	5

Future Work

Limitations of Current Work

- Many attributes originally had missing values, reducing usable feature information.

Suggestions for Improvement

- Collect or recover missing attribute metadata to improve interpretation.

Tasks to Complete Before End of Semester

- Train and tune Logistic Regression, J48, and deep neural network models.

REFERENCES

1. Rizkya, I., et al. "Analysis of Defective Causes in Coffee Product Using Decision Tree Approach." *IOP Conference Series: Materials Science and Engineering*. Vol. 851. No. 1. IOP Publishing, 2020.
2. Adobe. Coffee stock images. Adobe stock. Retrieved from https://stock.adobe.com/search?k=coffee+white+background (all coffee images were sourced from adobe stock)

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